powerful features, such as Language Integrated Query (LINQ), closures, and asynchronous programming constructs. The C# language is also designed in the open on GitHub, as is Microsoft's C# compiler, code-named Roslyn (https://github.com/dotnet/roslyn).

NOTE I use C# throughout this book and will highlight some of the newer features it provides, but I won't be teaching the language from scratch. If you want to learn C#, I recommend *C# in Depth*, 4th ed., by Jon Skeet (Manning, 2019), and *Code Like a Pro in C#,* by Jort Rodenburg (Manning, 2021).

One big advantage of ASP.NET Core and .NET 7 over .NET Framework is that they enable you to develop and run on any platform. With .NET 7 you can build and run the same application on Mac, Windows, and Linux, and even deploy to the cloud using tiny container deployments.

Built with containers in mind

Traditionally, web applications were deployed directly to a server or, more recently, to a virtual machine. Virtual machines allow operating systems to be installed in a layer of virtual hardware, abstracting away the underlying hardware. This approach has several advantages over direct installation, such as easy maintenance, deployment, and recovery. Unfortunately, virtual machines are also heavy, in terms of both file size and resource use.

This is where containers come in. Containers are far more lightweight and don't have the overhead of virtual machines. They're built in a series of layers and don't require you to boot a new operating system when starting a new one, so they're quick to start and great for quick provisioning. Containers (Docker in particular) are quickly becoming the go-to platform for building large, scalable systems.

Containers have never been a particularly attractive option for ASP.NET applications, but with ASP.NET Core, .NET 7, and Docker for Windows, all that is changing. A lightweight ASP.NET Core application running on the cross-platform .NET 7 framework is perfect for thin container deployments. You can learn more about your deployment options in chapter 27.